

SILISTRA, Bulgaria

WMO: 155500

Lat: 44.117N

Lon: 27.267E

Elev: 16

StdP: 101.13

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.3	-13.6	1.2	-10.5	-10.9	1.5	-7.7	16.0	2.7	14.0	-3.2	2.6	50	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	33.7	23.3	32.0	23.1	30.5	22.3	24.6	31.3	23.7	30.6	22.9	29.4	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.2	28.8	21.5	16.1	27.7	20.6	15.3	26.8	74.4	31.4	71.0	30.8	67.7	29.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	9.8	8.2	-13.2	36.6	3.3	2.5	-15.6	38.4	-17.5	39.9	-19.4	41.3	-21.8	43.1		
(5)			-13.6	26.1	3.2	1.1	-15.9	26.9	-17.7	27.5	-19.5	28.2	-21.8	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	0.3	2.7	7.6	13.1	18.4	22.2	24.6	24.4	19.2	13.2	7.7	2.1		
	DBStd	9.41	5.12	5.56	4.78	3.54	3.44	3.19	2.78	2.74	3.23	4.07	4.55	4.88		
	HDD10.0	990	301	208	105	12	1	0	0	0	17	97	249			
	HDD18.3	2556	558	438	334	161	43	6	0	1	28	163	319	504		
	CDD10.0	2089	1	4	29	104	260	367	453	446	276	118	28	3		
	CDD18.3	613	0	0	0	3	44	122	195	188	54	6	1	0		
	CDH23.3	5021	0	0	2	27	314	1028	1757	1595	279	19	1	0		
	CDH26.7	1797	0	0	0	3	64	354	713	608	54	1	0	0		
(14) Wind	WSAvg	2.5	2.6	3.0	3.4	2.7	2.5	2.2	2.1	1.9	2.2	2.2	2.4	2.7		
(15) Precipitation	PrecAvg	539	40	26	38	42	52	61	49	44	55	44	42	45		
	PrecMax	932	118	76	94	104	158	137	155	109	162	110	92	130		
	PrecMin	308	4	1	4	3	13	13	2	4	2	3	1	4		
	PrecStd	140	26	19	21	22	34	31	37	29	44	26	25	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.8	22.3	26.7	30.8	33.7	37.0	35.7	30.6	25.6	21.3	16.2		
		MCWB	10.4	11.6	13.6	17.5	20.4	23.7	23.3	23.5	20.9	19.2	15.9	12.1		
	2%	DB	11.3	14.7	19.7	23.0	28.3	31.8	34.3	33.3	28.2	23.3	18.7	13.2		
		MCWB	8.5	10.1	12.5	15.5	20.0	22.9	23.3	23.5	20.1	18.0	14.5	10.5		
	5%	DB	9.2	12.2	17.2	20.9	26.7	30.3	32.3	31.6	26.3	21.1	16.2	10.6		
		MCWB	6.9	8.5	11.2	14.5	19.4	22.2	23.1	23.0	19.4	16.9	13.3	8.6		
	10%	DB	7.2	10.0	15.0	19.0	24.9	28.7	30.5	30.1	24.7	19.3	14.0	8.6		
		MCWB	5.5	7.1	10.0	13.3	18.3	21.3	22.5	22.1	18.8	15.8	11.8	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.9	14.3	17.9	22.5	25.0	25.4	26.1	21.9	20.1	16.4	12.5		
		MCDB	13.9	16.9	21.3	24.9	28.1	31.6	32.5	31.7	28.4	24.5	20.6	15.5		
	2%	WB	8.6	10.2	12.7	16.2	20.9	23.8	24.4	24.7	20.8	18.3	14.9	10.6		
		MCDB	10.8	14.1	18.6	22.0	27.0	30.5	31.5	31.1	26.8	22.1	18.2	13.0		
	5%	WB	7.2	8.8	11.5	15.0	19.8	22.7	23.7	23.6	19.9	17.2	13.5	9.0		
		MCDB	9.0	12.0	16.9	20.0	25.5	29.4	30.8	30.2	25.4	20.6	15.9	10.6		
	10%	WB	5.5	7.2	10.2	13.9	18.7	21.8	22.9	22.7	19.1	15.9	12.0	7.1		
		MCDB	7.2	9.8	14.5	18.1	24.2	28.0	29.5	29.0	24.0	19.1	14.0	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.4	8.1	9.0	9.8	9.6	9.5	9.5	8.3	7.2	6.1	5.0		
		MCDBR	7.8	10.1	12.1	11.7	12.2	11.9	12.3	11.5	10.7	9.6	9.4	8.2		
	5% WB	MCWBR	5.8	6.7	6.7	6.2	6.3	5.5	5.1	5.0	5.0	5.7	6.1	6.3		
		MCWBR	7.5	10.0	11.4	10.6	11.1	11.1	11.0	10.7	9.8	9.0	8.9	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.379	0.414	0.464	0.438	0.444	0.456	0.469	0.437	0.398	0.371	0.350		
	taud		2.285	2.212	2.166	2.035	2.191	2.211	2.189	2.144	2.198	2.292	2.314	2.303		
	Ebn at Noon		751	790	813	800	837	831	815	788	780	762	717	706		
	Edh at Noon		86	110	131	161	142	140	142	143	126	100	84	77		
(44) All-Sky Solar Radiation	RadAvg		1.40	2.18	3.36	4.72	5.93	6.44	6.52	5.84	4.24	2.70	1.54	1.18		
	RadStd		0.20	0.31	0.34	0.50	0.41	0.45	0.40	0.32	0.32	0.29	0.20	0.15		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.47	N/A	N/A	N/A	N/A	N/A	-119	+87	+55

Nomenclature: See separate page